Muhammad Aamir

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Computer Scientist | AI/ML Researcher | Research Software Engineer | University Lecturer

Profile

Computer scientist, AI/ML researcher, research software engineer and university lecturer with a PhD in Computer Science and more than ten years of experience across research-intensive academia, applied research, industry collaboration and higher-education teaching.

My work sits at the intersection of **machine learning, computer vision, multimodal perception, software engineering and AI for Science**. I develop robust and reproducible computational methods for complex real-world data, with recent research focusing on 3D visual perception, depth estimation, metric reconstruction and large-scale biodiversity monitoring.

Alongside research, I teach undergraduate and postgraduate Computer Science and supervise research and software projects. My broader technical experience includes software architecture, APIs, client–server systems, databases, data pipelines, Docker, CI/CD, Digital Twins, AR/VR, NLP, knowledge graphs and scientific software engineering.

Research Interests

AI / ML: Deep learning; representation learning; multimodal learning; data-centric AI; robust and adaptive machine learning; AI for Science.

Computer Vision: 3D perception; depth estimation; object detection; image and video analysis; metric reconstruction; multimodal sensing; biological and environmental imaging.

Software Engineering: Reliable AI-enabled software; research software engineering; software architecture; testing; APIs; data pipelines; containerisation; CI/CD; reproducibility.

Applications: Biodiversity monitoring; wildlife conservation; biomedical imaging; industrial AI; Digital Twins; visual analytics; knowledge graphs.

Current Academic and Research Appointments

Department of Computer Science, University of Oxford — **Oxford, UK**

Senior Research Software Engineer (Machine Learning) — *Aug 2024–Present*

- Conduct AI/ML and computer-vision research within **MorphoCam**, a NERC–UKRI funded interdisciplinary programme linking Computer Science with the Wildlife Conservation Research Unit.
- Develop scalable and reproducible pipelines for camera-trap image/video analysis, multimodal perception, animal detection, depth estimation and metric measurement.
- Design modular research software spanning data preparation, model execution, evaluation, quality assurance, testing and deployment.
- Work with heterogeneous real-world data affected by domain shift, limited annotations, occlusion, class imbalance and environmental variation.
- Collaborate with computer scientists, machine-learning researchers, conservation scientists and field researchers to translate scientific questions into computational methods.
- Contribute to peer-reviewed publications, datasets, open research assets and technical dissemination.
- Mentor students and researchers in machine learning, computer vision, experiment design, software engineering and reproducibility.

Department of Computer Science, University of Oxford — **Oxford, UK**

Departmental Lecturer / Class Tutor (term-wise) — *Aug 2024–Present*

- Teach **Machine Learning, Databases and Data Visualisation** at undergraduate and postgraduate levels.
- Deliver classes, practical sessions and project supervision using authentic datasets and research-informed examples.
- Supervise MSc and undergraduate research/software projects from problem formulation through implementation, evaluation and academic writing.
- Contribute to assessment and MSc admissions, including application screening, candidate evaluation and interviews.
- Received a **Department Teaching Commendation in 2026** following excellent student feedback for clear and thorough Data Visualisation teaching.

Coventry University — **Coventry, UK**

Visiting Lecturer in Computer Science — *2026–Present*

- Deliver undergraduate and postgraduate Computer Science teaching.
- Connect software engineering, AI, data science and professional practice through current academic and industry examples.

Previous Academic and Professional Experience

University of the West of Scotland — **London, UK**
Associate Lecturer (part-time) — *May 2023–Jan 2026*

- Supervised MSc dissertations and BSc final-year projects in software engineering, computer science, information technology, machine learning and data science.
- Delivered lectures, practical sessions, project vivas, assessment and feedback.

University of Derby / Premtech Ltd — **Derby and Leicester, UK**
AI & Machine Learning Development Engineer — *Oct 2022–Jul 2024*

- Developed machine-learning and computer-vision methods for 2D/3D industrial asset recognition.
- Designed client–server software architectures and integrated machine learning with IoT, data pipelines and AR/VR systems.
- Worked with industrial stakeholders to elicit requirements, evaluate prototypes and transfer technical capability into operational practice.
- Contributed to applied research, technical dissemination and academic–industry knowledge exchange.

University of Derby / Bloc Digital — **Derby, UK**
KTP Associate – Software Engineer & Virtual Reality Specialist — *Oct 2020–Sep 2022*

- Designed and implemented a **Digital Twin platform** combining live data streams, analytics, 3D assets and immersive visualisation.
- Developed software using C#, Unity and Python, applying object-oriented design, modular architecture, APIs and data transformation.
- Coordinated work across academic and industrial teams and supported product development, staff training and technical knowledge transfer.
- Contributed to research on Digital Twin remote laboratories and online practical learning.

Universiti Tun Hussein Onn Malaysia — **Johor, Malaysia**
Lecturer (online) — *Aug 2020–Aug 2022*

- Taught undergraduate and master's Computer Science courses, developed learning materials and assessments, and supervised BSc, MSc and PhD research.
- Supported students with practical work, software implementation, evaluation, academic writing and viva preparation.

Universiti Tun Hussein Onn Malaysia — **Johor, Malaysia**
Postgraduate Research Assistant & Teaching Assistant — *Mar 2016–Jan 2020*

- Conducted research in machine learning, deep learning, software modularisation, optimisation, computer vision and text mining.
- Developed and evaluated machine-learning methods for high-dimensional and interdisciplinary datasets.
- Supported undergraduate and postgraduate teaching.

Orbit Software Solutions / Xululabs LLC — **Islamabad, Pakistan**
Data Scientist — *Feb 2015–Mar 2016*

- Developed NLP, social-media analytics, knowledge-graph and data-visualisation pipelines using Python, Java, Scala and R.
- Worked with MongoDB, relational databases, text processing, information extraction and relationship graphs.

Education

Universiti Tun Hussein Onn Malaysia — **Malaysia**
PhD in Computer Science / Information Technology — *2016–2020*
Thesis: *An Improved Contractive Autoencoder Based on Restricted Boltzmann Machine for Solving the Multiclass Classification Problem*. Research focus: deep representation learning, neural networks, optimisation and multiclass classification. **Supervisor:** Prof. Dr. Nazri bin Mohd Nawi.

City University of Science and Information Technology — **Peshawar, Pakistan**
Master of Computer Science — *2011–2013*
Thesis: *Web Application for Resume (PDF) Creation, Uploading and Text Extraction*. Focus: software development, document processing and text mining. **Supervisor:** Associate Prof. Dr. Rashid Naseem.

University of Peshawar — **Peshawar, Pakistan**
Bachelor of Science in Computer Science — *2008–2010*
Core study included programming, software engineering, databases, algorithms, artificial intelligence, data mining, probability and statistics, numerical analysis, networks and discrete mathematics.

Professional Development

Oxford Professional Education Group — **Oxford, UK**
Project Fundamentals Qualification — *2022*

Ashorne Hill Management College — **United Kingdom**
CMI Level 5 Certificate in Management and Leadership — *2021*

Selected Research Projects

NERC–UKRI **University of Oxford**
MorphoCam / Wildlife Conservation AI Programme *2024–Present*
Approximate project value: £702k. Computer vision, multimodal sensing, depth estimation, metric measurement, research software and reproducible AI for biodiversity monitoring.

Innovate UK / Industry **University of Derby / Premtech Ltd**
Asset Recognition and Capture *2022–2024*
Approximate project value: £76.5k. Industrial computer vision, 2D/3D asset recognition, software architecture, IoT integration and applied AI.

UKRI Knowledge Transfer Partnership / Industry **University of Derby / Bloc Digital**
Digital Twin Builder *2020–2022*
Approximate project portfolio value: £259k. Digital Twins, live-data integration, analytics, 3D assets, immersive visualisation and technical knowledge transfer.

Malaysian Ministry of Higher Education **Malaysia**
Health Surveillance of Rice Farms *2019–2020*
Drone-video and NDVI-based machine learning for crop-health monitoring.

Institute of Bioproduct Development **Malaysia**
Real-time Plant Species Classification *2018–2020*
Machine-learning classification using morphological, LC–MS and FT–IR measurements.

Selected Publications

2026: M. Aamir*, N. Muramatsu*, S. Shin*, M. Wijers, J. Zhong, X. Hou, A. Patel and A. Markham. "WildDepth: A Multimodal Dataset for 3D Wildlife Perception and Depth Estimation." Accepted at CVPR 2026; extended version at ECCV 2026.

2025: M. Aamir, M. Wijers, A. Loveridge and A. Markham. "A Robust Metric Distance and Height Estimation Pipeline for Wildlife Camera Trap Imagery." *Ecological Informatics*, 92, 103520.

2025: Q. Waseem, W. I. S. Wan Din and M. Aamir. "A Generalized Three-Tier Hybrid Model for Classifying Unseen IoT Devices in Smart Home Environments." *Scientific Reports*.

2024: A. Ashraf, N. M. Nawi, T. Shahzad, M. Aamir et al. "Dimension Reduction Using Dual-Featured Auto-Encoder for the Histological Classification of Human Lung Tissues." *IEEE Access*.

2024: F. Wahid, Y. Ma, R. Sheikh and M. Aamir. "An Enhanced Convolutional Neural Network Based on L1 Regularization for Segmentation of Coronary Arteries in X-Ray Angiograms." ICACS.

2022: C. Palmer, B. Roullier, M. Aamir et al. "Digital Twinning Remote Laboratories for Online Practical Learning." *Production & Manufacturing Research*.

2021: M. Aamir, N. M. Nawi, F. Wahid and H. Mahdin. "A Deep Contractive Autoencoder for Solving Multiclass Classification Problems." *Evolutionary Intelligence*.

Book: R. Ibrahim, N. A. Samudin, N. Senan and M. Aamir. *Object-Oriented Programming and Data Analysis using Java*. Penerbit UTHM, 2021.

Full record: 17+ journal articles, 6+ international conference papers and one academic book. Full publication list: Google Scholar.

Teaching and Supervision

Teaching areas: Software Engineering; Programming and Object-Oriented Development; Software Architecture and Design; Databases; Data Visualisation; Machine Learning; Artificial Intelligence; Data Science; Computer Vision; research, project and dissertation modules.

Teaching methods: Lectures; seminars; tutorials; laboratories; practical classes; live problem solving; authentic project work; online/blended teaching; formative feedback; research-informed case studies.

Assessment: Assessment design; marking; project vivas; dissertation/project assessment; formative feedback; evaluation of software and research artefacts.

Supervision: More than 30 MSc dissertations/projects, three PhD researchers, and numerous undergraduate and final-year research/software projects.

Student support: Academic mentoring; research-design guidance; project planning; software implementation; evaluation; academic writing; progression and career guidance.

Technical Skills

Programming: Python; C++; C#; Java; JavaScript; Scala; R; SQL; shell scripting.

AI / ML: PyTorch; TensorFlow; OpenCV; deep learning; CNNs; autoencoders; representation learning; classification; object detection; depth estimation; multimodal learning; model evaluation.

Data: Pandas; NumPy; SQL; relational databases; MongoDB/NoSQL; data preprocessing; feature engineering; visualisation; high-dimensional data analysis.

Software Engineering: Requirements analysis; object-oriented design; software architecture; APIs; client–server systems; testing; version control; Git/GitLab; Docker; Jenkins; CI/CD; reproducible environments.

Visual / Immersive: Unity; AR/VR; Digital Twins; 2D/3D asset recognition; image/video analysis; scientific visualisation.

NLP / Graphs: Text mining; document processing; social-media analytics; information extraction; knowledge graphs; relationship graphs.

Leadership and Academic Service

Leadership: Co-Chair, Mathematical, Physical and Life Sciences Division Research Staff Forum, University of Oxford.

Mentoring: Mentor students and researchers in machine learning, software engineering, research design, reproducibility and academic writing.

Peer review: Reviewer for *Neural Networks*, *Soft Computing*, *IEEE Access*, *PLOS ONE* and other international journals.

Conference service: Programme Committee member for international conferences in AI, soft computing, data mining and software engineering.

Membership: Member, IEEE Computer Society.

Awards and Recognition

University of Oxford Vice-Chancellor's Awards **Oxford, UK**
Making a Difference Globally Award – Winner *2026*
Recognised as part of the Wildlife Conservation Research Unit for research delivering global benefit to people and the planet.

Department of Computer Science, University of Oxford **Oxford, UK**
Department Teaching Commendation *2026*
Awarded following excellent student feedback for Data Visualisation classes, recognising clear explanations, thorough preparation and positive impact on learning.

Online Profiles

Personal Website Oxford Computer Science WildCRU Google Scholar ORCID GitHub